

Obesity and respiratory diseases: mechanisms, phenotypes, and clinical implications

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Obesity and Respiratory Diseases: Mechanisms, Phenotypes, and Clinical Implications

Running title: Obesity-Related Respiratory Disorders

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Abstract

Background:

Obesity has emerged as a global pandemic with profound implications for respiratory health. The complex interplay between excessive adiposity and pulmonary function encompasses mechanical, metabolic, and inflammatory pathways that significantly impact morbidity and mortality.

Objectives:

This comprehensive review synthesizes current evidence on obesity-related respiratory disorders, examining pathophysiological mechanisms, clinical phenotypes, epidemiological trends, and therapeutic considerations across the

spectrum of respiratory diseases. This review provides a novel integrative framework unifying mechanical, inflammatory, and metabolic mechanisms across all major obesity-related respiratory conditions, with emphasis on clinically actionable phenotyping and emerging pharmacological therapies.

Methods:

We conducted a narrative review of peer-reviewed literature (PubMed/MEDLINE, Embase, and Cochrane Database, January 2000 – December 2024, using pre-specified MeSH terms; study selection and quality assessment followed SANRA guidelines focusing on the relationship between obesity and respiratory disorders including asthma, obstructive sleep apnea (OSA), obesity hypoventilation syndrome (OHS), chronic obstructive pulmonary disease (COPD), respiratory infections, and lung cancer.

Results:

Obesity profoundly affects respiratory mechanics through a right-shift of the static volume-pressure relationship of the chest wall,

reducing FRC and ERV without a true restrictive pattern, altered respiratory muscle function, and systemic inflammation. Asthma prevalence increases with body mass index , demonstrating distinct phenotypes including atopic, insulin-resistant, dyslipidemic, and non-Th2 neutrophilic subtypes. OSA affects 22% of men and 17% of women, with obesity being the principal modifiable risk factor. OHS occurs in 8-20% of obese patients with sleep-disordered breathing, characterized by daytime hypercapnia (arising from impaired neuromuscular ventilatory drive and progressive nocturnal hypoventilation) and increased cardiovascular mortality. Paradoxically, obesity appears protective in advanced COPD by reducing FRC, thereby increasing inspiratory capacity and limiting dynamic pulmonary hyperinflation, and ARDS—the so-called obesity paradox—while conferring increased risk in COVID-19 pneumonitis. In lung cancer, obesity demonstrates complex relationships with risk and prognosis that vary by sex, smoking status, and disease stage.

Conclusions:

Obesity-related respiratory disorders represent a multifaceted clinical challenge requiring integrated multidisciplinary

approaches. Understanding distinct phenotypes and pathophysiological mechanisms is essential for personalized therapeutic strategies addressing both weight management and respiratory-specific interventions. Emerging pharmacological therapies including GLP-1 receptor agonists and dual GIP/GLP-1 agonists show particular promise in improving OSA severity and airway inflammation.

Keywords: Obesity, Respiratory diseases, Asthma phenotypes, Obstructive sleep apnea, Obesity hypoventilation syndrome, Obesity paradox, COVID-19, metabolic syndrome, GLP-1 receptor agonists, adipokines

1. Introduction

Obesity is a rapidly growing global health problem with significant consequences for respiratory health. In addition to its metabolic

and cardiovascular effects, excess adiposity alters pulmonary physiology through mechanical loading of the chest wall, reduced some lung volumes, impaired respiratory muscle function, and systemic inflammation. These changes contribute to increased respiratory symptoms, functional limitation, and adverse clinical outcomes across a broad range of pulmonary diseases.(1)

The spectrum of obesity-related respiratory pathology is broad. Obesity is not a direct cause of COPD or ARDS; however, it modifies their clinical course and through the obesity paradox may confer a survival advantage in severe disease. In contrast, obesity is the principal modifiable risk factor for OSA and OHS, is strongly associated with asthma — particularly non-atopic phenotypes — and worsens outcomes in respiratory infections including COVID-19 pneumonitis.

What distinguishes this review from prior works is its integrative, cross-disease mechanistic framework. Whereas previous reviews have addressed individual obesity-related respiratory conditions in isolation, we unify mechanical, adipokine-mediated, and

metabolic pathways across asthma, OSA, OHS, COPD, ARDS, and lung cancer. We further introduce updated evidence on GLP-1 receptor agonist therapies and their respiratory implications, and provide a phenotypic classification with direct clinical management guidance.

2- Methods

A systematic literature search was conducted in PubMed/MEDLINE, Embase, and the Cochrane Database of Systematic Reviews from January 2000 to December 2024. Search terms included: 'obesity AND respiratory diseases,' 'obstructive sleep apnea AND obesity,' 'obesity hypoventilation syndrome,' 'asthma AND obesity phenotype,' 'COPD AND obesity paradox,' 'ARDS AND obesity,' 'lung cancer AND BMI,' 'GLP-1 agonist AND respiratory function,' and related MeSH terms. Inclusion criteria: English-language peer-reviewed original articles, systematic reviews, and meta-analyses in human adult populations. Exclusion criteria: case reports, non-English publications, studies exclusively in pediatric populations (unless specifically relevant to mechanism), and pre-2000 publications except for seminal

foundational studies. Study quality was assessed using the SANRA (Scale for the Assessment of Narrative Review Articles) criteria. Where data were derived from animal or in vitro studies, this is explicitly indicated in the text. Evidence is presented hierarchically, with human clinical data prioritized over experimental findings.

3.Pathophysiological Mechanisms of Obesity-Related Respiratory Dysfunction (TABLE 1) (FIGURE 1)

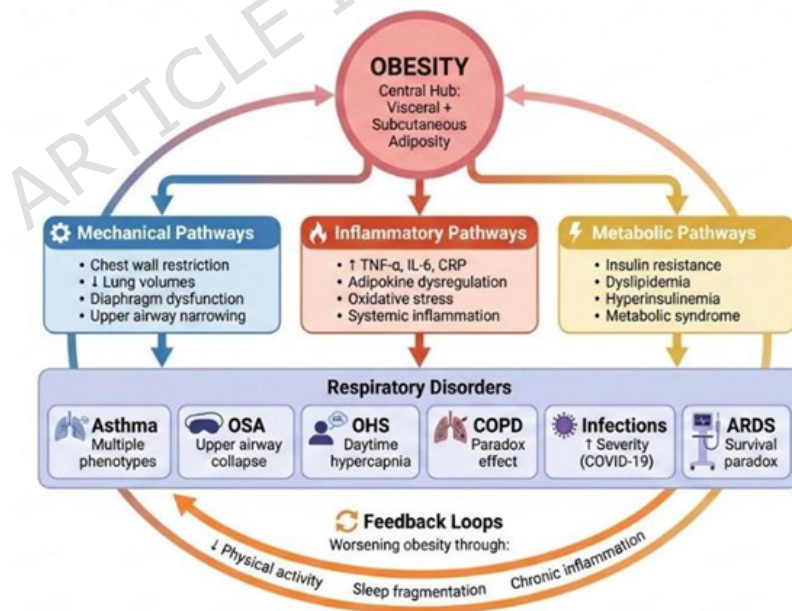


Figure 1. Comprehensive pathophysiological framework showing the bidirectional interactions between obesity and the respiratory system. Central obesity initiates three major pathways (mechanical, inflammatory, metabolic) that converge on multiple respiratory disorders. Bidirectional feedback loops perpetuate the cycle through reduced activity, sleep disruption, and chronic inflammation, creating a self-reinforcing pattern that worsens both obesity and respiratory dysfunction.

	Mechanism category	Specific effects	Clinical consequences	Key mediators
Mechanical Effects	Chest Wall	-Reduced compliance; -↑ work of breathing	Dyspnea on exertion; Respiratory muscle fatigue	Adipose tissue mass
	Lung Volumes	-↓ FRC, ↓ ERV, ↓ TLC	Small airway closure; V/Q mismatch; Hypoxemia	Abdominal adiposity
	Diaphragm	-Cephalad displacement -Reduced efficiency	Impaired ventilation; Orthopnea	Visceral fat; ↑ Intra-abdominal pressure
	Upper Airway	-Pharyngeal fat deposition	Increased OSA risk	Lateral pharyngeal wall fat
Inflammatory Pathways	Systemic	-Chronic low-grade inflammation	Airway inflammation; Hyperresponsiveness	TNF- α , IL-6, CRP
	Adipokines	-↑ Leptin, ↓ Adiponectin	Altered ventilatory control; Inflammation	Leptin, Adiponectin
Metabolic Dysfunction	Insulin Resistance	-Hyperinsulinemia -Impaired glucose metabolism	Airway hyperreactivity (especially in children)	Insulin, glucose
	Dyslipidemia	-↑ LDL, ↑ Triglycerides	Metabolic inflammation (male asthmatics)	LDL-C, triglycerides

Table 1: Pathophysiological Mechanisms of Obesity-Related Respiratory Dysfunction

Abbreviations: FRC=Functional Residual Capacity; ERV=Expiratory Reserve Volume; TLC=Total Lung Capacity; V/Q=Ventilation-Perfusion; OSA=Obstructive Sleep Apnea; TNF- α =Tumor Necrosis Factor-alpha; IL-6=Interleukin-6; CRP=C-Reactive Protein; LDL-C=Low-Density Lipoprotein Cholesterol

3.1 Mechanical Effects on Respiratory System

-Lung Volume Changes

Obesity-related alterations in the respiratory system occur at the morphological level and are closely intertwined with physiological disturbances. Structural changes lead to abnormal respiratory physiology, while physiological derangements can, in turn, promote or exacerbate morphological abnormalities. At the macroscopic level, excess fat deposition around the thoracic wall and abdominal cavity is a prominent feature.

Dysanapsis is another frequent finding in the lungs of has been demonstrated in overweight and obese children (1). It refers to a disproportion between airway size and lung volume, whereby the airways are relatively small for the amount of lung parenchyma. This imbalance is typically associated with preserved forced expiratory volume in one second (FEV_1), increased forced vital capacity (FVC), and a reduced FEV_1/FVC ratio(2) . In addition, airway wall thickening due to increased adipose tissue has been described. Experimental studies in porcine lungs demonstrated that adipose tissue extends along the conducting airways from the trachea to the subsegmental bronchi(3). Consistently, Elliott and colleagues showed that the area of adipose tissue within the

conducting airways of human lungs correlates positively with body mass index(4).

The primary mechanical consequence of obesity on the respiratory system is a right-shift of the static volume-pressure relationship of the chest wall. The weight of adipose tissue accumulated around — not within — the thoracic wall reduces its outward elastic recoil at low lung volumes and increases its inward elastic recoil at high lung volumes. This shifts the relaxation volume of the respiratory system — the functional residual capacity (FRC) — toward a lower lung volume. Since residual volume (RV) does not change appreciably in obesity, expiratory reserve volume (ERV) is consequently reduced. This is not a static volume effect on the lung itself, but a consequence of the altered chest wall mechanics. The chest wall static compliance is roughly normal or only mildly reduced, but pleural pressure is less negative at every lung volume, increasing the load on the inspiratory muscles.(5,6). As the degree of obesity increases, ERV declines in an exponential manner and shows an inverse relationship with other static lung

volumes and capacities(5,7). A comparable pattern has also been demonstrated for end-expiratory lung volume.(8)

Total lung capacity (TLC) is generally maintained within normal limits; however, in individuals with morbid obesity, TLC may decrease by as much as 20%(5), producing a restrictive ventilatory pattern. Accordingly, when a reduced TLC is observed in patients with a body mass index up to 40 kg/m², clinicians should consider the presence of an alternative or coexisting respiratory disorder, as obesity alone is unlikely to account for such marked impairment in lung volumes.

Alterations in static lung volumes occur much more commonly in individuals with an abdominal (android) pattern of obesity than in those with a gynoid fat distribution, largely because of the greater mechanical effect on diaphragmatic motion (9). Vital capacity (VC), forced vital capacity (FVC), and forced expiratory volume in one second (FEV₁) are typically mildly reduced in people with obesity, which results in a normal or near-normal FEV₁/FVC ratio(6,7,10)

In contrast, the mid-expiratory flow rate (FEF_{25-75%} of FVC) may be markedly decreased (7), reflects reduced lung volumes (FVC)

with proportionate decrements in FEV1, maintaining a normal FEV1/FVC ratio — thus representing a volume effect rather than true small airways obstruction. The possibility of airway remodeling secondary to adipokine-mediated inflammation is retained but appropriately qualified as speculative. Diffusing capacity is usually preserved or even increased, consistent with the enhanced pulmonary blood flow characteristic of obesity(11).

-Chest Wall Mechanics

The effects of obesity on lung volumes are largely driven by the influence of chest compliance. In individuals with obesity, chest wall compliance is reduced because excess adipose tissue over the chest and abdomen mechanically restricts chest wall expansion and limits diaphragmatic movement(11,12)

Furthermore, the obesity-related reduction in functional residual capacity (FRC) may increase alveolar surface tension, which further impairs lung compliance(13). Nevertheless, evidence suggests that decreased lung compliance plays a smaller role compared with the dominant effect of reduced thoracic (chest wall) compliance(14).

-Diaphragmatic Function

Accumulation of adipose tissue around the thoracic and abdominal cavities raises pleural pressure, thereby restricting rib cage movement and diaphragmatic excursion. Consequently, the work of breathing increases, while the elastic energy available for passive exhalation declines.

Studies have reported variable findings regarding respiratory muscle strength: some demonstrate values within normal limits, while others show increased strength in morbidly obese patients(15,16). In contrast, respiratory muscle endurance—the capacity to sustain repeated contractions against resistance—is consistently reduced compared with normal-weight individuals.(17)

Both inspiratory and expiratory times appear shortened in obesity, contributing to an increased respiratory rate from approximately 10–12 breaths per minute in normal subjects to 15.3–21 breaths per minute in obese individuals (18). This tachypnea compensates

for the reduced tidal volume, resulting in normal or increased minute ventilation, particularly in the supine position.(19,20)

-Upper Airway Effects

Airway narrowing with a consequent rise in resistance is common in obesity, even when the FEV₁/FVC ratio remains preserved(21) . Body plethysmography has shown a progressive increase in airway resistance with higher BMI, while forced oscillation studies demonstrated nearly double the resistance in obese individuals compared with controls(22,23).

The increased thickness of the airway smooth muscle layer in obesity has been attributed to persistently low functional residual capacity (FRC), which causes continuous smooth muscle shortening and adaptation for greater strength at shorter lengths(24).

In addition, obese subjects exhibited increased ΔX_{rs} (the difference between inspiratory and expiratory reactance), independent of reduced FRC, indicating the presence of additional mechanisms contributing to airflow limitation.(25)

The mechanisms underlying airway narrowing in obesity can be broadly divided into mechanical factors and airway hyperresponsiveness related to a pro-inflammatory state that will be discussed in the following section .

3.2 Inflammatory and Metabolic Pathways

-Systemic Inflammation

Obesity is characterized by a systemic pro-inflammatory state, marked by elevated leukocytes, cytokines, and chemokines (26), which contributes to several complications, including cardiovascular disease (27), glucose intolerance (28), and malignancy (29). This inflammatory milieu may also underlie airway hyperresponsiveness in obese individuals. Since airway eosinophil and neutrophil levels are not significantly elevated, the inflammation appears primarily systemic rather than localized to the airways (21).

Serum tumor necrosis factor (TNF)- α levels are increased in obesity. TNF- α receptors are present on airway smooth muscle,

and in vitro exposure enhances smooth muscle contractility (30). Leptin, elevated in obesity, exerts pro-inflammatory effects including promotion of Th2 polarization, enhancement of eosinophil survival, and upregulation of IL-4 and IL-13 in airway epithelial cells. These effects promote bronchial hyperresponsiveness and airway remodeling. (31).

-Adipokine Dysregulation

Adiponectin, another hormone produced by adipose tissue, is typically reduced in obese individuals (32). Because adiponectin inhibits TNF- α and exerts anti-inflammatory effects by suppressing NF- κ B signaling, as well as inhibiting mast cell activation, decreased adiponectin levels may enhance airway smooth muscle contractility. (33). Notably, bariatric surgery reduces airway hyperresponsiveness and is associated with increased serum adiponectin and decreased leptin levels (34).

The leptin/adiponectin imbalance characteristic of obesity thus creates a pro-asthmatic milieu.

Obesity-related hypertrophy and hyperplasia of airway smooth muscle are mainly observed with coexisting asthma, where high mobility group box protein 1 (HMGB1) plays a key role. Intracellular HMGB1 regulates gene transcription, whereas extracellular HMGB1 promotes inflammatory and immune responses. Its expression is significantly increased in the adipose tissue of obese individuals (35) and has been shown to enhance airway smooth muscle thickness in murine asthma models (36). This effect is mediated through stimulation of smooth muscle cell proliferation via RAGE receptor activation and downstream ERK/NF- κ B signaling pathways (37). Additionally, HMGB1 promotes epithelial-mesenchymal transition by reducing tight junction protein expression (38). Collectively, molecular pathways involving HMGB1, RAGE, and ERK/NF- κ B have been implicated in adipokine-mediated airway remodeling; however, these mechanisms are primarily characterized in experimental (murine and in vitro) models and still require validation in human studies.

In addition to these mechanisms, other molecular pathways likely contribute to airway hyperresponsiveness in obesity and may be clarified by future research (39).

3.3 Metabolic Dysfunction

-Insulin Resistance and Hyperinsulinemia

Insulin resistance in obesity leads to compensatory hyperinsulinemia, which exerts direct pro-contractile and pro-fibrotic effects on the airways. Insulin stimulates airway smooth muscle proliferation and enhances bronchial hyperresponsiveness by increasing intracellular calcium signaling. Additionally, insulin resistance impairs nitric oxide-mediated vasodilation, contributing to pulmonary microvascular dysfunction. Hyperinsulinemia also upregulates insulin-like growth factor-1 (IGF-1), promoting extracellular matrix deposition and airway remodeling. These mechanisms are particularly relevant in obese asthma phenotypes and may exacerbate airflow limitation and reduce responsiveness to inhaled corticosteroids.(40)

-Dyslipidemia

Obesity-related dyslipidemia, characterized by elevated triglycerides, increased LDL cholesterol, and reduced HDL cholesterol, contributes to chronic lung disease through lipotoxicity and inflammation. Accumulation of oxidized lipids in pulmonary tissues activates macrophages and induces the release of pro-inflammatory mediators. Cholesterol crystals can stimulate the NLRP3 inflammasome, leading to IL-1 β -mediated airway inflammation. Reduced HDL impairs its normal anti-oxidant and anti-inflammatory functions, further enhancing oxidative injury. These processes promote endothelial dysfunction, impaired alveolar repair, and accelerated decline in lung function, particularly in COPD and obesity-associated asthma.(41)

-Oxidative Stress

Obesity is associated with increased production of reactive oxygen species (ROS) due to mitochondrial dysfunction,

chronic inflammation, and reduced antioxidant capacity. In the lungs, excessive ROS cause epithelial and endothelial injury, inactivation of antiproteases, and increased mucus production. Oxidative stress activates redox-sensitive transcription factors such as NF- κ B, perpetuating inflammatory cascades. It also contributes to corticosteroid resistance by impairing histone deacetylase-2 (HDAC2) activity, a key mechanism in severe asthma and COPD. Persistent oxidative injury accelerates airway remodeling, emphysematous changes, and disease progression in chronic lung disorders.(42)

3.4 Effects on Ventilatory Control

To maintain adequate tidal volume and alveolar ventilation for normal gas exchange, obese individuals require a greater neuromuscular respiratory central drive to sustain the inspiratory muscles against increased elastic and resistive workloads. In most obese individuals, this compensatory drive is sufficient. However, a subset — due to genetic or non-genetic factors including low growth hormone levels, reduced circulating leptin, or central leptin resistance — fails to mount an adequate ventilatory

response. This leads to progressive hypoventilation and CO₂ retention, initially during REM sleep, subsequently during NREM sleep, and ultimately when awake, culminating in established primary OHS. (18).

4- Obesity and Asthma: A Complex Relationship

4.1 Epidemiological Evidence

The strongest association between obesity and respiratory disease has been observed with asthma (43-45). Adult-onset asthma is more common in individuals with BMI >25 kg/m² compared with those of lower BMI (46). Asthma prevalence increases with rising BMI, with a 12.5% increase in individuals with BMI 25–29.9 kg/m² and a markedly higher prevalence in those with BMI ≥50 kg/m². Asthma severity is also greater in obese individuals, reflecting a dose-dependent relationship between obesity and asthma. Obesity further increases the likelihood of a persistent and severe asthma phenotype (47).

This association is independent of sociodemographic factors, physical activity, and dietary habits. One study identified a linear relationship between asthma severity and BMI(48).

Two meta-analyses published in 2013 and 2019 examined the obese asthma phenotype and confirmed a positive association between asthma and childhood overweight or obesity (49,50). The first meta-analysis, including six prospective studies, reported a 20% increase in asthma incidence among overweight children and a twofold higher risk in obese children, with a stronger association in boys before puberty (49). The second analysis, which included 18 prospective studies, found a 20% increased risk in overweight children and a 40% increase in obese children (50).

Body mass index (BMI) is the most commonly used measure to assess weight status in adults and children because of its simplicity. In children, overweight is defined as BMI above the 85th percentile and obesity as BMI above the 95th percentile (51). However, BMI does not differentiate between fat and lean mass or between subcutaneous and visceral fat, although fat distribution is a stronger predictor of asthma development (52).

4.2 Distinct Phenotypes of Obese Asthma(TABLE 2) (FIGURE 2)

Figure 2: Obesity-Asthma Phenotypes - Distinct Phenotypes of Obesity-Related Asthma

 Atopic Phenotype	 Insulin-Resistant	 Dyslipidemic	 Non-Th2 Neutrophilic
Features: • ↑ IgE • ↑ Eosinophils • Early onset • Allergic triggers Treatment: ✓ Good ICS response	Features: • Hyperinsulinemia • Later-onset disease (typically after age 12) • Glucose intolerance Treatment: ✗ Poor ICS response 💊 Insulin sensitizers	Features: • ↑ LDL & Triglycerides • Adult males • Metabolic syndrome • Non-Th2 inflammation Treatment: ⚠ Variable ICS response 💊 Lipid management	Features: • ↑ Neutrophils • Late onset • Severe symptoms • Frequent exacerbations Treatment: ✗ Poor ICS response 💊 IL-17 biologics

Clinical Implications

Recognition of distinct phenotypes enables precision medicine approaches. Rather than uniform treatment, clinicians can tailor therapy based on inflammatory profile, metabolic status, and biomarkers. This phenotype-directed strategy improves outcomes and reduces unnecessary medication use.

Table 2: Phenotypic Classification of Obesity-Related Asthma

Phenotype	Characteristics	Predominant mechanisms	Biomarkers	Treatment response	Demographics
Atopic Obesity-Related Asthma	• Early onset (childhood/adolescence) • Allergic sensitization present • Eosinophilic airway inflammation • Positive allergy testing	• Th2-mediated inflammation • Amplified allergic responses • Mechanical effects of obesity	↑ IgE ↑ Eosinophils ↑ FeNO - Positive skin prick tests	- Good ICS response - Standard therapy effective - Anti-IgE biologics helpful	Both genders equally affected; Children and adults
Insulin-Resistant Phenotype	• later-onset disease (typically after age 12) • Hyperinsulinemia present • Glucose intolerance or impaired glucose metabolism • Metabolic syndrome features	• Insulin-mediated airway hyperreactivity • Non-Th2 inflammatory pathways • Metabolic dysfunction	↑ Insulin levels ↑ HOMA-IR - Impaired glucose tolerance - Normal/low eosinophils	- Poor ICS response - May benefit from insulin sensitizers - Weight loss crucial	later-onset disease (typically after age 12)
Dyslipidemic Phenotype	• Adult onset (typically >30 years) • Elevated lipid levels • Male predominance (significant finding) • Often with metabolic syndrome	• Metabolic inflammatory derangement • Oxidative stress pathways • Non-Th2 mediated inflammation	↑ LDL cholesterol ↑ Triglycerides - Dyslipidemia panel - Variable eosinophils	- Variable ICS response - Lipid management may help - Weight reduction beneficial	Predominantly males; Adult population
Non-Th2 Neutrophilic Phenotype	• Late onset (adult) • Severe symptoms and frequent exacerbations • Poor asthma control • Neutrophilic sputum	• Neutrophilic airway inflammation • Systemic inflammatory state • Mechanical respiratory restriction	↑ Neutrophils ↑ Systemic inflammatory markers - Low eosinophils - Low/normal FeNO	- Poor ICS response - High-dose ICS may be needed - May need IL-17 or other biologics	Adults; Both genders affected

Abbreviations: Th2 = T-helper type 2; IgE = Immunoglobulin E; FeNO = Fractional Exhaled Nitric Oxide; ICS = Inhaled Corticosteroids; HOMA-IR = Homeostatic Model Assessment for Insulin Resistance; LDL = Low-Density Lipoprotein; IL-17 = Interleukin-17

-Atopic Phenotype

Early-onset asthma EOA resembles allergic, eosinophilic Th2-driven asthma and responds to corticosteroids. EOA is accompanied by greater airway obstruction and positively correlated with BMI. The patients had greater (airway hyperresponsiveness) AHR and the risk was 4–6 fold higher for hospitalization.(53)

-Insulin-Resistant Phenotype

Typically late-onset and associated with metabolic dysfunction, this phenotype is marked by airway hyperresponsiveness and remodeling driven by hyperinsulinemia and insulin resistance. Inflammation is often non-eosinophilic, and patients may exhibit reduced responsiveness to corticosteroids.(54)

-Dyslipidemic Phenotype

Associated with abnormal lipid profiles, this phenotype involves lipotoxic airway inflammation mediated by oxidized lipids and macrophage activation. Oxidative stress and inflammasome

activation contribute to airflow limitation and poorer asthma control.(41)

-Non-Th2 Inflammatory Phenotype

Late-onset nonallergic (LONA) asthma is characterized by Th1/Th17 responses and neutrophilic inflammation and poorly responds to corticosteroid treatment. It affected more women than men, was characterized by lower AHR, more persistent airway remodeling, poorer symptom control and more frequent exacerbations. (55)

4.3 Mechanistic Considerations

Mechanical and inflammatory mechanisms converge in obesity-related asthma. Obese individuals breathe just above RV favoring the occurrence of tidal expiratory flow limitation at rest and near to the closing volume promoting the functional closure of dependent small airways (both phenomena occurring mainly in the supine position) which further impair airways function. Supporting this mechanism, interventions that increase lung

volume, such as continuous positive airway pressure, reduce airway reactivity in experimental models and asthma patients (56).

Visceral adipose tissue releases metabolic mediators and adipokines, especially leptin, which are elevated in obese asthmatics and correlate with airway reactivity. Although systemic inflammatory markers are increased, this phenotype is not always linked to greater airway inflammation, suggesting direct adipokine effects on airway structure and function.

Overall, mechanical restriction and adipose-derived inflammatory signaling explain the obesity-asthma relationship, though behavioral, hormonal, and genetic factors may also contribute (57).

4.4 Clinical Implications

Asthma in obese patients is associated with more severe symptoms, poorer disease control, and increased exacerbations despite often modest airway inflammation. Reduced lung volumes,

mechanical restriction, and obesity-related comorbidities contribute to disproportionate dyspnea and reduced quality of life. Obese asthma is frequently less responsive to standard asthma therapies, particularly inhaled corticosteroids, highlighting the need for phenotype-based management. Addressing obesity and associated conditions through weight reduction and multidisciplinary care can significantly improve asthma outcomes and treatment responsiveness.

5. Obesity Hypoventilation Syndrome

5.1 Definition and Diagnostic Criteria

Obesity hypoventilation syndrome (OHS) is characterized by obesity (body mass index ≥ 30 kg/m²), sleep-disordered breathing, and daytime hypercapnia (arterial PaCO₂ ≥ 45 mmHg at sea level) during wakefulness, without an alternative neuromuscular, mechanical, or metabolic cause of hypoventilation (58). About 90% of patients with OHS have obstructive sleep apnoea (OSA), defined

by an apnoea-hypopnoea index (AHI) ≥ 5 events/h, and nearly 70% present with severe OSA (AHI ≥ 30 events/h) (59).

5.2 Epidemiology and Risk Factors

Obesity hypoventilation syndrome occurs in about 8–20% of obese patients with sleep-disordered breathing and is strongly linked to severe obesity (BMI ≥ 35 kg/m²). Risk rises with central adiposity, male sex, older age, and coexisting obstructive sleep apnea. Other contributors include reduced lung volumes, leptin resistance, physical inactivity, and cardiopulmonary disease. OHS is often underrecognized, frequently misdiagnosed as COPD or heart failure, resulting in delayed treatment and higher morbidity and mortality (60).

5.3 Pathophysiology

-Mechanical Load

In OHS, the increased mechanical load from thoraco-abdominal adiposity results in elevated airway resistance and respiratory muscle inefficiency. These changes increase airway resistance, may cause dependent airway closure leading to ventilation-perfusion mismatch and basal atelectasis, and — by favoring tidal expiratory flow limitation — may generate intrinsic positive end-expiratory pressure [PEEPi]. Patients with obesity hypoventilation syndrome show greater impairment in respiratory mechanics and muscle function than similarly obese individuals without OHS, resulting in increased work of breathing and higher ventilatory demand (61).

-Impaired Ventilatory Drive

Most obese individuals maintain normal CO₂ levels by increasing respiratory drive to offset the higher breathing workload. When this compensation fails, hypoventilation occurs, initially during REM sleep, when muscle atonia increases reliance on diaphragmatic function and central drive. In obesity hypoventilation syndrome, mechanical diaphragmatic restriction and impaired central drive cause REM-related hypoventilation.

Leptin resistance contributes by reducing ventilatory stimulation and promoting cardiometabolic dysfunction. A typical OHS phenotype includes morbid obesity, severe mechanical impairment, marked hypercapnia, and prominent REM hypoventilation, often responsive to noninvasive ventilation (62).

-Sleep-Disordered Breathing

Obesity-related physiological changes worsen during supine sleep, explaining the high prevalence of obstructive sleep apnea in obese individuals. Fat accumulation around the upper airway and reduced lung volumes narrow the pharynx and increase airway collapsibility, promoting obstruction(63). Nocturnal rostral fluid shift from the legs to the neck further constricts the airway, especially in fluid overload states.

In obesity hypoventilation syndrome, obstructive events are often prolonged and poorly compensated because of impaired central respiratory drive, causing nocturnal carbon dioxide retention and daytime hypoventilation (64). A distinct OHS phenotype—marked by predominant OSA without REM-related hypoventilation or

respiratory muscle dysfunction, usually in less severely obese patients—generally responds well to continuous positive airway pressure therapy(65).

5.4 Diagnosis and Severity Grading

The American Academy of Sleep Medicine defines adult sleep hypoventilation as $\text{PaCO}_2 > 55$ mmHg for over 10 minutes, or a rise > 10 mmHg from awake levels to > 50 mmHg for the same duration. Although this reflects daytime abnormality, nocturnal polysomnography or respiratory polygraphy is required to assess sleep-related hypoventilation and distinguish obstructive from non-obstructive patterns, guiding appropriate positive airway pressure therapy.

The European Respiratory Society also proposed an OHS severity classification, identifying early disease by elevated serum bicarbonate and/or sleep hypoventilation, and severe disease by daytime hypercapnia with cardiovascular and metabolic comorbidities (62).

5.5 Treatment and Outcomes

First-line treatment includes positive airway pressure therapy, with continuous positive airway pressure for patients with predominant obstructive sleep apnea and noninvasive ventilation for those with persistent hypercapnia. Weight reduction through lifestyle modification or bariatric surgery is the only disease-modifying intervention and significantly improves ventilatory function and survival. Untreated OHS is associated with high cardiovascular morbidity and mortality; however, early recognition and appropriate therapy markedly improve gas exchange, quality of life, and long-term outcomes.

6. Obstructive Sleep Apnea

6.1 Epidemiology and Risk Factors

Obstructive sleep apnea (OSA), defined by an apnea-hypopnea index ≥ 5 , showed a mean prevalence of 22% (9–37%) in men and 17% (4–50%) in women across eleven epidemiological studies (1993–2013). OSA with excessive daytime sleepiness affected 6%

of men and 4% of women. Prevalence rose over time, reaching 37% in men and 50% in women in later studies (2008–2013). Overall, OSA is more common in men and increases with age and obesity(66).

Excessive daytime sleepiness is considered the main symptom of obstructive sleep apnea (OSA), yet only some individuals with AHI >5 report it, and one study found no association between sleep apnea and daytime sleepiness in women(66).

Obesity is a major risk factor for snoring and OSA, with most affected patients being overweight(67). Weight reduction through caloric restriction or bariatric surgery decreases OSA severity, while higher BMI increases disease risk; a 10% weight gain raises OSA risk six-fold(68). Other risk factors include male sex, older age, craniofacial structure, increased neck circumference, and comorbidities such as insulin resistance and hypothyroidism. Weight gain worsens OSA, whereas weight loss significantly reduces disease burden.

6.2 Pathophysiology

Obesity contributes to OSA mainly through two mechanisms: 1) fat deposition in the anterior neck physically narrows the airway during sleep, and 2) reduced FRC decreases tracheal tug and upper airways stiffness, increasing their collapsibility (69).

OSA-related poor sleep may further raise BMI by lowering serum leptin, an appetite suppressant(70), which can increase appetite and promote weight gain(71).

6.3 Cardiovascular and Metabolic Consequences

Obesity-related OSA causes intermittent hypoxia, sympathetic activation, and systemic inflammation, raising risks of hypertension, coronary disease, heart failure, stroke, and arrhythmias. It also promotes insulin resistance, dyslipidemia, and type 2 diabetes, with obesity and OSA together amplifying cardiometabolic risk(72).

6.4 Treatment Considerations

CPAP therapy remains the gold standard for OSA. Weight reduction through lifestyle intervention, bariatric surgery, and emerging pharmacological therapies effectively reduces apnea-hypopnea index [AHI], oxygen desaturation burden, and daytime symptom burden, and reduces cardiovascular risk. Adjunctive therapies include positional therapy, oral appliances in selected patients, and management of contributing comorbidities. A multidisciplinary approach integrating sleep medicine, respiratory care, and metabolic management is essential for optimal outcomes.(73)

7. Obesity and Chronic Obstructive Pulmonary Disease: The Obesity Paradox

7.1 Prevalence and Relationship

The interaction between obesity and COPD is less dose responsive. Although the prevalence of COPD does increase with increasing BMI , the severity of COPD adds complexity. In a study from the Netherlands, among patients with Global Initiative for

Chronic Obstructive Lung Disease (GOLD) stage 1 and 2 COPD the prevalence of obesity was 16–24%, whereas in patients with GOLD stage 4 COPD the prevalence of obesity was 6%. This was compared with a national prevalence of 10% (74, 75)

7.2 The Obesity Paradox

Studies of mechanisms explaining the interaction between COPD and obesity are scarce, aside from sleep-disordered breathing. Both impact pulmonary mechanics and ventilatory demand, likely amplifying each other. An “obesity paradox” is reported in COPD: increased BMI appears protective against all-cause mortality. Annual FEV1 decline in men was lower in obese versus non-obese COPD patients.(76). This reflects the link between low BMI and mortality. The paradox varies with COPD phenotypes: in advanced COPD, obesity is protective, while in mild COPD, it may increase cardiovascular risk, a greater mortality contributor than low BMI.(77)

7.3 Potential Mechanisms

Obesity may protect in COPD through the “obesity paradox.” Greater nutritional reserves buffer catabolic stress during exacerbations, and preserved skeletal and respiratory muscle supports ventilation, exercise capacity, and resilience in advanced disease. Earlier presentation, treatment tolerance, and intensive management may also help.(78)

Clinically, obesity poses a dual challenge: it can worsen dyspnea, reduce lung volumes, and limit activity, yet may improve prognosis in advanced COPD by lowering FRC, so increasing inspiratory capacity and limiting dynamic pulmonary hyperinflation.(78)

7.4 Clinical Implications

The interaction between obesity and COPD has important clinical consequences. Clinical management should therefore focus on body composition rather than BMI alone, aiming to preserve muscle mass while avoiding excess fat accumulation. Individualized strategies incorporating pulmonary rehabilitation, nutritional optimization, and careful weight management are

essential to balance symptom burden with potential protective effects of obesity in COPD.

8. Obesity, Respiratory Infections, and ARDS

8.1 Viral Respiratory Infections

Although obesity induces a systemic pro-inflammatory state, its link to respiratory infections was unclear before COVID-19 due to conflicting data. Animal studies of influenza showed obesity weakens antiviral responses (79,80), aligning with observations made during H1N1 influenza that obesity increases disease severity (81-83), ICU admission, and death (84) . COVID-19 confirmed obesity as a poor prognostic factor, raising risks of pneumonitis (85), hospitalization, and mortality(86).

Multiple mechanisms link obesity to COVID-19, notably its effect on the angiotensin-converting enzyme 2 (ACE2) receptor. ACE2 serves as the entry point for SARS-CoV-2 via its spike protein(87). Present on lung epithelial cells and adipocytes, ACE2 expression is increased in obese individuals, facilitating viral entry(88).

Adipocytes can serve as reservoirs and replication sites for SARS-CoV-2(89), potentially prolonging viral shedding, as seen with influenza(90). Obesity also amplifies COVID-19 severity via interleukin-6 (IL-6) (91), a key risk factor, since adipose tissue secretes IL-6 (92). Additionally, leptin, overexpressed in obesity, promotes pro-inflammatory cytokines like TNF- α and IL-6, contributing to the cytokine storm linked to morbidity and mortality(93,94). Other molecular mechanisms are still under investigation

In summary, obesity creates a pro-inflammatory state that amplifies COVID-19 inflammation. Obese individuals also more commonly have chronic conditions—hypertension, cardiovascular disease, chronic kidney disease, and type 2 diabetes—that further increase COVID-19 morbidity and mortality.(95)

8.2 Acute Respiratory Distress Syndrome (ARDS)

Evidence linking obesity to bacterial pneumonia remains unclear. Some studies report higher self-reported respiratory infections (96) and hospitalization risk (97), yet many highlight an obesity survival paradox. A meta-analysis of 12 studies (1.5 million patients) (98) and subsequent studies (99,100), showed obesity increases infection risk but lowers mortality. Mechanistic explanations for differences between viral and bacterial infections are lacking. While many theories have been speculated, none are yet supported by compelling evidence (101). Obesity is also an independent risk factor for acute respiratory distress syndrome.(102)

Obesity raises the risk of ARDS during severe infection, trauma, or sepsis through increased systemic inflammation and reduced lung reserve. Mechanical changes, such as lower functional residual capacity and less chest wall expandability, complicate ventilation, often requiring higher airway pressures and careful lung-protective strategies. Despite these challenges, obesity does not consistently worsen short-term ARDS outcomes.

8.3 The Obesity Survival Paradox in ARDS

Obesity has been linked to equal or better survival compared with normal BMI, illustrating the “obesity survival paradox,” where higher BMI ICU patients may have lower mortality(103-105). Explanations include greater nutritional reserves, altered inflammation, earlier ICU admission, and more intensive care. Reduced chest wall expandability may also protect lungs during ventilation. Though not fully understood, this paradox underscores the importance of tailoring ARDS management to physiology and body composition rather than BMI alone.

9. Obesity and Lung Cancer

9.1 Epidemiological Relationships

Excess adiposity may promote cancer by increasing inflammation, angiogenesis, metastatic potential, and apoptosis evasion (106).

Obesity is linked to higher mortality in several cancers, including colorectal, liver, gallbladder, pancreatic, and kidney (107). However, an “obesity paradox” exists in lung cancer: meta-analyses show high BMI independently predicts lower risk, improved treatment outcomes, and longer overall survival (108) , contrary to the usual harmful effects of excess fat.

9.2 Variability by Demographics

The association between obesity and lung cancer varies by sex, age, smoking status, and ethnicity (109). Stronger inverse associations have been observed in current and former smokers, whereas relationships are less consistent in never-smokers.(110). Sex-specific differences may reflect hormonal influences, fat distribution patterns, and differences in immune responses(111) .Age-related changes in body composition and metabolic health further modify lung cancer risk and prognosis.(112)

9.3 Potential Mechanisms

The link between obesity and lung cancer involves interactions among adipose tissue, inflammation, hormones, and metabolism. Adipose tissue acts as an endocrine organ, releasing pro-inflammatory cytokines and adipokines that promote tumor initiation. Insulin resistance and hyperinsulinemia drive carcinogenesis via insulin and IGF-1 pathways (113) . Altered sex hormone metabolism, especially increased estrogen, and obesity-related oxidative stress further contribute to DNA damage and genomic instability, collectively increasing lung cancer risk.(113)

The “obesity paradox” refers to the surprising observation that, in some populations, overweight or obese individuals have lower mortality than those with normal weight.(114).

In cancer, proposed mechanisms include protective effects of adipokines with anti-inflammatory and immunomodulatory roles (115) , greater metabolic reserves aiding tolerance to treatments(116) , and obesity-related genetic and molecular changes—such as altered insulin-like growth factor IGF signaling, chronic inflammation, PI3K/AKT/mTOR pathway modulation, and estrogen metabolism—that may influence tumor biology and

therapy response. Metabolic reprogramming, hypoxia-driven angiogenesis, and epigenetic modifications may also affect outcomes (117,118) . The complex interplay between obesity, cancer biology, and treatment response requires further investigation.

9.4 Clinical Implications

The interaction between obesity and lung cancer underscores the importance of considering body composition, metabolic health, and immune status in risk assessment and treatment planning. While obesity may confer protective effects in certain contexts, it can also complicate cancer management through comorbidities and altered pharmacokinetics.

Personalized therapeutic strategies incorporating nutritional assessment, metabolic optimization, and careful dosing of anticancer therapies are essential to improve outcomes in obese patients with lung cancer.

10. Clinical Assessment and Management of Respiratory Disease in Obesity

10.1 Comprehensive Respiratory Evaluation in Obese Patients (TABLE 3)

Table 3 :Comparative Epidemiology and Clinical Features of Obesity-Related Respiratory Disorders

Condition	Prevalence in obese population	Gender distribution	Key risk factors beyond obesity	Primary clinical features	Major complications
Asthma	11% in obese individuals vs 7% in normal weight (BMI <25)	Equal overall, but varies by phenotype	• Age (increases with age) • Neck circumference >40cm (men), >34cm (women) • Craniofacial anatomy abnormalities	• Loud snoring • Witnessed apneas • Excessive daytime sleepiness (variable)	• Hypertension • Cardiovascular disease • Metabolic syndrome
Obstructive sleep apnea	• Affects approximately 40-70% of obese individuals • Prevalence increases in proportion to BMI	• More common in males • Gender difference decreases after menopause	• Advancing age • Increased neck circumference (>40 cm in men, >34 cm in women) • Craniofacial and upper airway anatomical abnormalities	• Loud habitual snoring • Witnessed apneas during sleep • Excessive daytime sleepiness (variable severity)	• Systemic hypertension • Cardiovascular and cerebrovascular disease • Metabolic syndrome and insulin resistance
Obesity hypoventilation syndrome	8-20% of obese patients referred to	Equal gender distribution	• Severe obesity (BMI ≥ 40 kg/m²)	• Daytime hypercapnia (PaCO₂ ≥ 45 mmHg)	• Cor pulmonale • Acute/chronic

	sleep centers	(unlike OSA)	• Severe OSA (AHI ≥ 30 events/hour) • Elevated serum bicarbonate	• Dyspnea at rest or with minimal exertion • Marked hypersomnolence	respiratory failure • Significantly increased mortality
COPD (mild-moderate)	16-24% in GOLD stage 1-2 COPD	Variable; depends on smoking patterns	• Smoking history (primary risk factor) • Occupational exposures • Alpha-1 antitrypsin deficiency	• Obesity paradox effect • Increased cardiovascular risk • Better survival than lean COPD patients	• Cardiovascular events (major cause of mortality) • Acute exacerbations
COPD (severe)	Only ~6% in GOLD stage 4 COPD (inverse relationship)	• Variable • Reflects historical smoking exposure patterns	• Long-term heavy smoking • Frequent or severe exacerbations • Chronic systemic inflammation	• Strong protective effect of obesity • Better survival compared to underweight patients • Slower FEV₁ decline	• Respiratory failure • Cachexia (if not obese - poor prognostic sign)
COVID-19 pneumonitis	High-risk subgroup with significantly worse outcomes	Equal gender risk	• Age >65 years • Multiple comorbidities • Metabolic syndrome components	• Severe pneumonia • Rapid progression to ARDS • Prolonged hospitalization	• ICU admission • Mechanical ventilation requirement • Significantly increased mortality
ARDS	Increased risk of developing ARDS	Equal gender distribution	• Sepsis • Trauma/major surgery • Aspiration pneumonia	• Severe hypoxemia • Survival paradox (better outcomes once ARDS develops) • Bilateral infiltrates	• Multi-organ failure • Death (but lower mortality vs lean patients - paradox)

Abbreviations: BMI = Body Mass Index; AHI = Apnea-Hypopnea Index; ICS = Inhaled Corticosteroids; OSA = Obstructive Sleep Apnea; OHS = Obesity Hypoventilation Syndrome; PaCO₂ = Partial Pressure of Carbon Dioxide; COPD = Chronic Obstructive Pulmonary Disease; GOLD = Global Initiative for Chronic Obstructive Lung Disease; FEV₁ = Forced Expiratory Volume in 1 second; ARDS = Acute Respiratory Distress Syndrome; ICU = Intensive Care Unit

A thorough respiratory evaluation in obese patients should begin with a detailed history focusing on dyspnea, sleep-related symptoms, exercise intolerance, exacerbation frequency, and obesity-associated comorbidities such as OSA, OHS, GERD, and cardiovascular disease. Physical examination should assess body fat distribution, neck circumference, chest wall mechanics, signs of hypoventilation, and features of right heart strain.

Pulmonary function testing commonly reveals a reduced functional residual capacity and expiratory reserve volume. This pattern does not constitute a true restrictive ventilatory defect which would require a proportional reduction in TLC. Clinicians should recognize this ERV/FRC-predominant pattern as distinct from classical restriction, avoiding misclassification. Spirometry may reveal proportionate reductions in FEV₁ and FVC with a normal FEV₁/FVC ratio, reflecting the reduced lung volumes rather than airflow obstruction.. Arterial blood gas analysis is

essential when hypoventilation is suspected, particularly to detect hypercapnia and chronic respiratory acidosis.

Sleep studies, including polysomnography or respiratory polygraphy, are critical for diagnosing sleep-disordered breathing, while exercise testing helps distinguish deconditioning from intrinsic pulmonary or cardiac limitations.(119)_.Diagnostic Approach to Obesity-Related Respiratory Disorders was shown in TABLE 4

Table 4: Diagnostic Approach to Obesity-Related Respiratory Disorders

Clinical presentation	First line investigations	Confirmatory tests	Screening criteria	Differential diagnoses
Dyspnea on exertion	Spirometry Chest X-ray Pulse oximetry	• Full PFTs with DLCO • Arterial blood gas (ABG) • Cardiopulmonary exercise testing (CPET)	All obese patients (BMI ≥ 30) presenting with dyspnea	• Cardiac dysfunction/heart failure • Deconditioning • Anemia
Suspected OSA	STOP-BANG questionnaire Neck circumference Overnight SpO ₂	• Polysomnography (gold standard) • Home sleep apnea testing (HSAT)	• Snoring + witnessed apneas • Excessive daytime sleepiness • BMI > 30 kg/m²	• Central sleep apnea • Sleep movement disorders • Narcolepsy
Suspected OHS	Arterial blood gas (ABG) Serum bicarbonate Pulse oximetry	• Sleep study with CO₂ monitoring • Nocturnal hypoventilation assessment • Echocardiogram (assess for cor pulmonale)	• BMI ≥ 30 kg/m² • Serum HCO₃⁻ ≥ 27 mEq/L (97% NPV if < 27) • Daytime hypoxemia	• Neuromuscular disease • Chest wall disorders • Severe COPD

Asthma + obesity	Spirometry with bronchodilator Peak flow variability	•Methacholine challenge test •FeNO measurement •Blood eosinophils •Metabolic panel (insulin, lipids, glucose)	•Wheezing and dyspnea •Variable airflow obstruction •BMI >30 kg/m²	•Vocal cord dysfunction •Cardiac asthma •COPD/asthma overlap
Obesity + COPD	Spirometry (post-bronchodilator)) Chest X-ray Detailed smoking history	•Full PFTs (lung volumes, DLCO) •CT chest (assess emphysema) •6-minute walk test •Body composition analysis	•Smoking history ≥10 pack-years •Progressive dyspnea •Fixed airflow obstruction (FEV₁/FVC <0.70)	•Bronchiectasis •Interstitial lung disease •Cardiac disease

Abbreviations: PFTs = Pulmonary Function Tests; DLCO = Diffusing Capacity for Carbon Monoxide; ABG = Arterial Blood Gas; CPET = Cardiopulmonary Exercise Testing; SpO₂ = Oxygen Saturation; STOP-BANG = Screening questionnaire for OSA; HSAT = Home Sleep Apnea Testing; HCO₃⁻ = Bicarbonate; NPV = Negative Predictive Value; FeNO = Fractional Exhaled Nitric Oxide; FEV₁ = Forced Expiratory Volume in 1 second; FVC = Forced Vital Capacity

10.2 Integrated Management Approaches TABLE 5

Table 5: Management Strategies for Obesity-Related Respiratory Disorders

Condition	Weight management	Respiratory specific therapy	Combined approach benefits	Monitoring parameters
Obese asthma	•Dietary modification (target 5-10% weight loss) •Structured exercise program •Behavioral modification •Consider bariatric surgery if BMI >40 kg/m²	•ICS ± LABA (may need higher doses) •Phenotype-directed biologic therapy •Address insulin resistance/dyslipidemia	Improved asthma control Reduced exacerbations Better treatment response	•Spirometry every 3-6 months •Asthma Control Test scores •Exacerbation frequency •Weight trajectory

		pidemia if present •Treat comorbidities		
OSA	•10% weight loss → approximately 26% reduction in AHI •Lifestyle modification programs •Bariatric surgery (highly effective for severe obesity) - Respiratory-Specific Therapy: •CPAP or BiPAP therapy (first-line) •Oral appliances (mild OSA) •Positional therapy •Upper airway surgery (selected cases)	•CPAP or BiPAP therapy (first-line) •Oral appliances (mild OSA) •Positional therapy •Upper airway surgery (selected cases)	•CPAP + weight loss superior to either alone May allow CPAP discontinuation with significant weight loss - Improved cardiovascular outcomes	•AHI on therapy (download data) •CPAP adherence (hours/night) •Daytime symptoms (ESS score) •Blood pressure and cardiovascular parameters
OHS	•Intensive weight loss programs •Bariatric surgery (strong consideration) •Nutritional support and monitoring	•BiPAP or volume-assured pressure support •High-intensity CPAP (if primarily OSA) •Oxygen supplementation if needed •Consider non-invasive ventilation	- Improved daytime blood gases - Reduced hospitalization - Better cardiovascular outcomes - Reduced mortality	•Awake ABG (PaCO₂, pH) •Nocturnal hypoventilation monitoring •Serum bicarbonate levels •Echocardiogram for pulmonary hypertension
COPD+ Obesity	•Mild COPD: Encourage weight loss (reduce CV risk) •Severe COPD: Cautious approach - maintain muscle mass •Focus on balanced nutrition and body composition	•Standard COPD therapy (bronchodilators, ICS if indicated) •Pulmonary rehabilitation (critical) •Manage cardiovascular comorbidities •Smoking cessation	- Better in mild disease (↓ CV risk) - Preserve muscle mass in severe disease - Individualized based on severity	•Spirometry (FEV₁ trajectory) •Exercise capacity (6MWT) •Body composition analysis •Exacerbation rate and severity

Post-COVID/ARDS Recovery	• Gradual, monitored weight management • Nutritional rehabilitation • Physical reconditioning programs • Address post-intensive care syndrome	• Pulmonary rehabilitation (essential) • Oxygen therapy if persistent hypoxemia • Treatment of pulmonary fibrosis if present • Psychological support	Improved functional recovery Better long-term outcomes Enhanced quality of life	• Oxygen saturation (rest and exertion) • 6-minute walk distance • Pulmonary imaging (CT if indicated) • Quality of life assessments
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Abbreviations: ICS = Inhaled Corticosteroids; LABA = Long-Acting Beta-Agonist; BMI = Body Mass Index; AHI = Apnea-Hypopnea Index; CPAP = Continuous Positive Airway Pressure; BiPAP = Bilevel Positive Airway Pressure; ESS = Epworth Sleepiness Scale; ABG = Arterial Blood Gas; PaCO₂ = Partial Pressure of Carbon Dioxide; CV = Cardiovascular; 6MWT = 6-Minute Walk Test; FEV₁ = Forced Expiratory Volume in 1 second

Management of respiratory disease in obesity requires an integrated, patient-centered approach. Weight reduction through lifestyle interventions or bariatric surgery is a cornerstone of therapy and can substantially improve respiratory mechanics, gas exchange, and disease control. Condition-specific treatments, such as inhaled therapies for asthma or COPD and positive airway pressure for OSA and OHS, should be optimized based on disease phenotype. Multidisciplinary care involving pulmonologists, sleep specialists, nutritionists, physiotherapists, and psychologists

improves adherence and long-term outcomes. Patient education is essential to enhance understanding of disease mechanisms, promote lifestyle modification, and improve treatment compliance.

10.3 Special Considerations

Obese patients require careful perioperative respiratory assessment due to increased risks of hypoventilation, atelectasis, and postoperative pulmonary complications. During pregnancy, obesity increases the risk of asthma exacerbations and sleep-disordered breathing, necessitating close monitoring and individualized management. In pediatric populations, obesity-related respiratory disorders such as asthma and OSA may have long-term developmental consequences, highlighting the importance of early identification, family-centered interventions, and prevention strategies.(61)

10.4 Emerging Pharmacological Therapies for Obesity and Respiratory Outcomes

GLP-1 receptor agonists and dual GIP/GLP-1 receptor agonists have emerged as transformative therapies for obesity with direct and indirect benefits on respiratory function. Semaglutide (STEP-1 trial) achieved mean weight reduction of approximately 14.9%, with secondary improvements in OSA severity, exercise capacity, and systemic inflammation. (120) Liraglutide demonstrated significant reductions in AHI in the SCALE Sleep Apnea trial. (121) Tirzepatide (SURMOUNT-1 trial) achieved weight reductions of up to 22.5%,(122) and the SURMOUNT-OSA trial demonstrated significant AHI reductions (approximately 29 events/hour in the non-PAP group) independent of weight loss, suggesting GLP-1 receptor-mediated effects on upper airway muscle tone and inflammation. (123)

In asthma, GLP-1 receptor agonists may reduce exacerbation frequency through both weight-mediated and direct anti-inflammatory mechanisms, though large prospective trials are pending. These agents represent a paradigm shift in the management of obesity-related respiratory disorders and should be considered in the integrated treatment plan of obese patients with significant respiratory comorbidities.(124)

11. Discussion

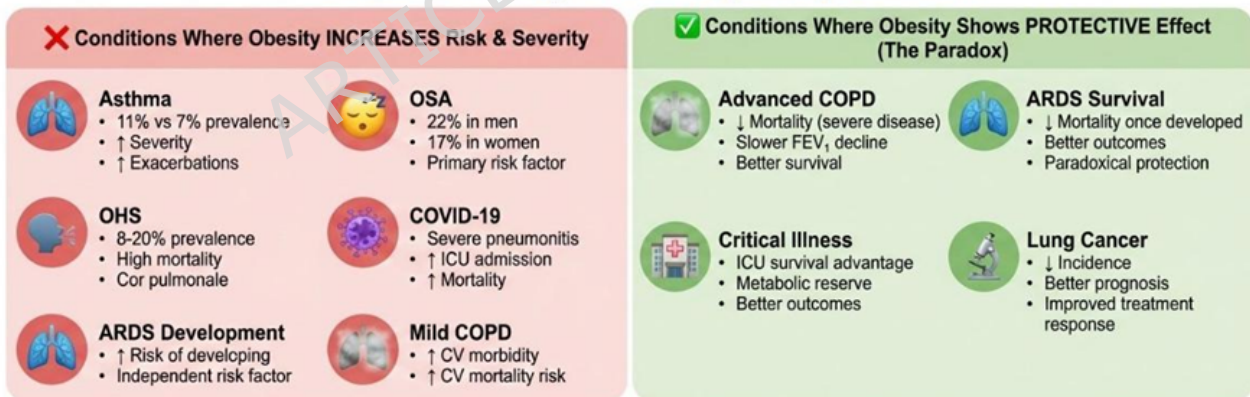
This review has synthesized the evidence for obesity's impact on the full spectrum of respiratory diseases, with a focus on mechanistic pathways, clinical phenotyping, and emerging therapies. The unifying theme is that obesity generates a chronic state of mechanical disadvantage and pro-inflammatory dysregulation that affects every level of the respiratory system — from upper airway patency (OSA) to peripheral airway function (asthma) to ventilatory control (OHS) to alveolar mechanics (ARDS).

A key insight is that not all effects of obesity are detrimental: the obesity paradox in COPD and ARDS reflects genuine mechanistic benefits **as shown in figure 3** -- FRC reduction improving dynamic hyperinflation in COPD, and preserved metabolic reserve potentially limiting organ failure in ARDS. Clinicians should resist reducing obesity to a uniformly negative variable and instead recognize context-specific effects.

The adipokine axis — particularly the leptin/adiponectin imbalance — offers a promising therapeutic target. Beyond

mechanical and pharmacological weight reduction, targeting adipokine signaling may provide respiratory benefits independent of absolute weight loss. GLP-1 receptor agonists may represent the first class of agents capable of simultaneously addressing the metabolic, inflammatory, and mechanical dimensions of obesity-related respiratory disease

Figure 3: The Obesity Paradox in Respiratory Disease - Differential Effects of Obesity Across Respiratory Conditions



Understanding the Paradox: Key Insights

1. **Disease Stage Matters:** Obesity harmful in mild COPD (↑ cardiovascular risk) but protective in severe COPD (metabolic reserve hypothesis).
2. **Development vs. Progression:** Obesity increases risk of developing ARDS but paradoxically improves survival once ARDS occurs.
3. **Metabolic Reserve Hypothesis:** Greater fat stores provide catabolic protection during critical illness, allowing patients to better withstand metabolic stress.
4. **Clinical Implications:** Weight management strategies must be individualized based on disease type, severity, and stage rather than applying universal recommendations.

Figure 3. The obesity paradox demonstrates differential effects of obesity across respiratory conditions. While obesity clearly increases risk for asthma, OSA, OHS, and COVID-19 severity, it paradoxically shows protective effects in advanced COPD, ARDS survival, critical illness, and lung cancer outcomes. This complexity highlights the need for nuanced, condition-specific and stage-specific management strategies rather than universal weight loss recommendations for all obese patients with respiratory disease.

12. Future Directions and Research Priorities

Future research should prioritize refined phenotyping of obesity-related respiratory disorders to support precision medicine approaches, as reliance on body mass index alone fails to capture the heterogeneity of adiposity, fat distribution, and metabolic status influencing disease expression. Standardization of obesity phenotyping—distinguishing visceral from subcutaneous fat and accounting for metabolic health—along with integration of clinical, physiological, metabolic, inflammatory, and imaging biomarkers, may improve identification of disease subtypes and prediction of therapeutic response. Further mechanistic studies are needed to elucidate the complex pathways linking obesity to respiratory dysfunction, including adipokine signaling, metabolic inflammation, ventilatory control abnormalities, airway and parenchymal remodeling, and the gut-lung microbiome axis. These efforts may also clarify paradoxical observations, such as the apparent survival benefit of obesity in advanced COPD and ARDS.

Therapeutic innovation remains a key priority, particularly through targeted anti-inflammatory and metabolic interventions

and combination strategies that integrate pharmacologic treatments with structured weight reduction. In this context, large prospective trials of GLP-1 and dual GLP-1/GIP agonists with respiratory outcomes as primary endpoints are warranted, alongside sex-stratified analyses to better define differential responses. Bariatric surgery continues to show significant benefits in asthma, OSA, and ventilatory mechanics, though further research is needed to optimize patient selection and long-term outcomes. Additionally, pediatric cohort studies are essential to determine whether early-life factors, such as dysanapsis in obese children, predispose to adult respiratory disease. At the population level, public health strategies focusing on obesity prevention, early intervention, multidisciplinary care, and integration of respiratory outcomes into obesity management programs will be critical to reducing disease burden.

13. Limitations

This narrative review has several inherent limitations. First, as a narrative rather than systematic review, selection bias cannot be fully excluded despite the SANRA-guided approach. Second, the

epidemiological literature is predominantly cross-sectional, limiting causal inference. Third, heterogeneity in obesity definitions (BMI cut-offs, body fat distribution measures) across studies complicates comparisons. Fourth, most studies have been conducted in Western adult populations; data from pediatric and non-Western populations are limited. Fifth, evidence hierarchies are not always available — some mechanistic statements are derived primarily from experimental models and require human validation

14. Conclusions

Obesity-related respiratory disorders represent a multifaceted clinical challenge requiring integrated, multidisciplinary approaches. A mechanistic understanding of the distinct phenotypes — grounded in the adipokine axis, chest wall mechanics, and ventilatory control physiology — is essential for personalized therapeutic strategies. Weight reduction remains the cornerstone of management, but emerging pharmacological therapies including GLP-1 receptor agonists offer new possibilities

for addressing both the metabolic and respiratory dimensions of obesity. Future work should focus on sex-specific phenotyping, microbiome interactions, and the translational potential of adipokine-targeted therapies.

15.Conflicts of Interest:

The authors declare that there are no conflicts of interest.

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ABBREVIATIONS :

OSA obstructive sleep apnea
 OHS obesity hypoventilation syndrome
 COPD chronic obstructive pulmonary disease
 ARDS acute respiratory distress syndrome
 FEV1 forced expiratory volume in one second
 FVC forced vital capacity
 ERV expiratory reserve volume
 FRC functional residual capacity
 TLC Total lung capacity
 VC Vital capacity
 BMI body mass index
 HMGB1 high mobility group box protein 1
 TNF α tumor necrosis factor α
 IGF-1 insulin-like growth factor-1
 ROS reactive oxygen species
 HDAC2 impairing histone deacetylase-2
 AHR airway hyperresponsiveness
 EOA Early-onset asthma
 LONA Late-onset nonallergic asthma
 AHI an apnoea-hypopnoea index
 GOLD Global Initiative for Chronic Obstructive Lung Disease
 ACE2 angiotensin-converting enzyme
 GIP/GLP-1 glucose-dependent insulintropic polypeptide and glucagon-like peptide-1

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